

[IMAGE: WakeCap Logo - Company letterhead placeholder]

POWER SOLUTIONS

Operations Selection Guide

Site Assessment & Solution Selection for Field Deployment

[IMAGE: Decision flowchart graphic or product lineup comparison image]

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INTERNAL USE - OPERATIONS DEPARTMENT

Target: Project Engineers | Site Engineers | Operations Managers

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1. Quick Selection Guide

Use this section for rapid solution selection based on your equipment load requirements.

1.1 Selection by Load

Match your total system power consumption to the appropriate Power Solution:

Total System Load	Recommended Solution	Typical Equipment	Backup Time*
≤ 8W	WakeCap Power 30 (WP-30)	WakeCap Gateway only	48+ hours
9W - 15W	WakeCap Power 60 (WP-60)	Single bullet camera, small sensor	48+ hours
16W - 100W	WakeCap Power 400 (WP-400T)	PTZ camera, multiple bullets, gateway + cameras	48+ hours
101W - 350W	WakeCap Power 900 (WP-900T)	Multiple PTZ cameras, comm equipment	48+ hours

* Backup time assumes zero solar input (worst case). Actual runtime varies with solar conditions.

CRITICAL

Always calculate TOTAL SYSTEM POWER, not just device power. System power includes the device, PoE injectors, routers, voltage converters, and all auxiliary equipment.

1.2 Equipment Power Reference

Standard power consumption for common equipment:

Equipment Type	Device Power	System Power*	Notes
WakeCap Gateway	~5W	~8W	Includes power conversion losses
Bullet Camera (standard)	15W	35W	Includes PoE injector, router
PTZ Camera (Type 1)	42W	70W	Mid-range PTZ with PoE
PTZ Camera (Type 2)	62W	90W	High-end PTZ with heater
Network Router/Switch	5-15W	—	Add to system total
PoE Injector	3-8W	—	Conversion losses

* System Power = Device + PoE + Router + Conversion Losses. Always use System Power for selection.

2. Load Calculation Fundamentals

Proper load calculation is essential for selecting the correct power solution. Follow this systematic approach.

2.1 Calculation Order

Power system sizing follows this sequence:

1. LOAD - Calculate total continuous power consumption
2. BATTERY - Size battery bank for required backup duration
3. SOLAR - Size solar array to recharge battery daily
4. MPPT - Select controller to handle solar current
5. AUXILIARY - Add protection, cabling, mounting

2.2 Standard Design Parameters

All WakeCap Power Solutions are designed using these parameters:

Parameter	Symbol	Value	Description
Backup Time	T	48 hours	Required autonomy without solar input
Depth of Discharge	DoD	70%	Usable battery capacity
System Efficiency	η_{sys}	90%	Wiring and general losses
Converter Efficiency	η_{conv}	80%	DC-DC conversion losses
Solar Derating	η_{solar}	75%	Dust, misalignment, temperature
MPPT Efficiency	η_{mppt}	95%	Charge controller efficiency
Peak Sun Hours	PSH	6.5 hours	Saudi Arabia average

2.3 Battery Capacity Formula

Required battery capacity (Ah) for a given load:

$$\text{Battery (Ah)} = (T \times \text{Load}) / (\text{DoD} \times \eta_{sys} \times \eta_{conv} \times V)$$

Where: T = 48h, DoD = 0.70, η_{sys} = 0.90, η_{conv} = 0.80, V = System Voltage

2.4 Example Calculation

Calculate the solution for a single PTZ Camera (Type 1) system:

- System Load: 70W (42W camera + PoE + router)
- System Voltage: 12V
- Required Battery = $(48 \times 70) / (0.70 \times 0.90 \times 0.80 \times 12)$
- Required Battery = $3360 / 6.048 = 556 \text{ Ah}$

RESULT

A 70W load requires approximately 556Ah battery capacity. The WP-400T (400Ah) is insufficient; select WP-900T (900Ah @ 24V = equivalent to ~450Ah usable at 12V reference, but provides 24V system) for margin. Alternatively, reduce load or accept shorter backup time.

2.5 Load Capacity Summary

Maximum continuous loads per solution (for 48-hour backup):

Solution	Battery Capacity	Max Load (48h backup)	Max Load (24h backup)
WP-30	30Ah / 384Wh	~8W	~16W
WP-60	60Ah / 768Wh	~15W	~30W
WP-400T	400Ah / 4,800Wh	~100W	~200W
WP-900T	900Ah @ 24V / 21,600Wh	~350W	~700W

3. Product Specifications Comparison

3.1 Power System Specifications

Specification	WP-30	WP-60	WP-400T	WP-900T
System Voltage	12V DC	12V DC	12V DC	24V DC
Battery Capacity	30Ah	60Ah	400Ah (2×200)	900Ah (6×150)
Battery Energy	384Wh	768Wh	4,800Wh	21,600Wh
Battery Type	LiFePO4	LiFePO4	GEL Lead Acid	HT Lead Acid
Solar Power	80W	100W	450W	1,305W
Solar Panels	2 × 40W	2 × 50W	3 × 150W	3 × 435W
MPPT Controller	Integrated	10A	30A	60A EPEVER
MPPT Efficiency	>95%	96.5%	[TBD]	[TBD]
Output Voltage	12V ±0.5V	10.5-12.6V	12V DC	24V DC
Max Output Current	3A	10A	[TBD]	[TBD]

3.2 Physical Specifications

Specification	WP-30	WP-60	WP-400T	WP-900T
Total Weight	~6 kg	23 kg	525 kg	1,300 kg
Form Factor	Pole/Wall Mount	Bracket Mount	Trailer	Trailer
Mast Height	N/A	N/A	6.5m (pneumatic)	9m (manual)
Transport Size (mm)	780×530×200	940×780×240	2000×1400×2300	2320×1495×2350
Deployed Size (mm)	780×530×varies	940×780×varies	2000×2250×6500	[TBD]
Trailer Type	N/A	N/A	Single axle	Single axle
Tow Hitch	N/A	N/A	Ball hitch	50mm ball hitch
Tire Size	N/A	N/A	165/70R13	15"

3.3 Environmental Ratings

Specification	WP-30	WP-60	WP-400T	WP-900T
IP Rating	IP66	IP66	IP65	IP65
Operating Temp	-20°C to +50°C	-30°C to +85°C	[TBD]	[TBD]
Charge Temp (Low)	-10°C	-30°C	[TBD]	[TBD]
Discharge Temp (Low)	-20°C	-30°C	[TBD]	[TBD]

Wind Rating (Panel)	2400Pa	60m/s (200kg/m ²)	[TBD]	100 km/h
Snow Load	5400Pa	[TBD]	[TBD]	[TBD]

BATTERY CHEMISTRY NOTE

LiFePO₄ batteries (WP-30, WP-60) offer better cold weather performance and longer cycle life. GEL/Lead Acid batteries (WP-400T, WP-900T) are more cost-effective for high-capacity applications but require more frequent replacement.

4. Site Assessment Checklist

Complete this assessment before selecting and deploying a Power Solution.

4.1 Load Requirements

- ☐ List all equipment to be powered:
 - Equipment 1: _____ Power: _____ W
 - Equipment 2: _____ Power: _____ W
 - Equipment 3: _____ Power: _____ W
 - PoE Injectors / Converters: _____ W
 - Network Equipment: _____ W
- ☐ TOTAL SYSTEM LOAD: _____ W
- ☐ Required backup time without solar: _____ hours (standard: 48h)

4.2 Site Access

- ☐ Access road available for:
 - Light vehicle (WP-30, WP-60): ☐ Yes ☐ No
 - Pickup truck with trailer (WP-400T): ☐ Yes ☐ No
 - Heavy truck with trailer (WP-900T): ☐ Yes ☐ No
- ☐ Site accessible during deployment period: ☐ Yes ☐ No
- ☐ Crane or lifting equipment available (if needed): ☐ Yes ☐ No

4.3 Ground Conditions

- ☐ Ground type: ☐ Compacted soil ☐ Sand ☐ Gravel ☐ Concrete ☐ Other: _____
- ☐ Ground is level (within 5°): ☐ Yes ☐ No
- ☐ Ground can support trailer weight: ☐ Yes ☐ No ☐ N/A
- ☐ Anchor points available (for trailers): ☐ Yes ☐ No ☐ N/A

4.4 Mounting Structure (WP-30, WP-60 only)

- ☐ Existing pole available: ☐ Yes ☐ No
- ☐ Pole diameter: _____ mm (WP-30 clamps fit Ø150mm)
- ☐ Pole structural capacity verified: ☐ Yes ☐ No
- ☐ Wall mounting possible: ☐ Yes ☐ No

4.5 Solar Exposure

- ☐ Clear southern sky exposure: ☐ Yes ☐ Partial ☐ No
- ☐ Shading from structures/equipment: ☐ None ☐ Morning ☐ Afternoon ☐ Full day
- ☐ Estimated Peak Sun Hours: _____ hours/day (Saudi average: 6.5h)
- ☐ Seasonal variation considered: ☐ Yes ☐ No

5. Environmental Considerations

Saudi Arabia deployment environments require special attention to these factors.

5.1 Desert Environment

- High ambient temperatures (up to 50°C+)
- Solar panel derating at high temperatures (expect 10-15% reduction)
- Battery capacity reduction at extreme temperatures
- Dust accumulation on solar panels - plan monthly cleaning
- Sand/dust ingress - verify IP rating adequate for location

HIGH TEMPERATURE

At temperatures above 45°C, expect reduced charging efficiency and shorter battery runtime. LiFePO4 batteries (WP-30, WP-60) handle heat better than Lead Acid (WP-400T, WP-900T). Consider shading the battery enclosure if possible.

5.2 Coastal/Humid Environment

- Salt air accelerates corrosion on exposed metal
- Higher humidity reduces solar panel efficiency slightly
- Verify all connections are sealed and protected
- More frequent maintenance inspections recommended
- Consider additional corrosion protection for trailer frames

5.3 Wind Considerations

Maximum wind ratings by product:

- WP-30/WP-60: Solar panels rated for 60 m/s (216 km/h)
- WP-400T/WP-900T: Mast should be lowered in high winds
- WP-900T: System rated for 100 km/h wind speed

MAST SAFETY

Always lower trailer masts before wind speeds exceed 50 km/h. Extended masts act as significant wind loads and can cause trailer instability or structural damage.

5.4 Security Considerations

- Site fencing recommended for all trailer deployments
- Wheel locks and hitch locks for theft deterrence
- Consider GPS tracking for high-value trailer units

- Remote monitoring via cellular (if available) for intrusion alerts

6. Installation Requirements

6.1 Personnel Requirements

Solution	Min. Personnel	Skill Level	Special Equipment
WP-30	1 technician	Basic electrical	Ladder, hand tools
WP-60	1 technician	Basic electrical	Ladder, hand tools, drill
WP-400T	2 technicians	Electrical + mechanical	Tow vehicle, level, hand tools
WP-900T	2-3 technicians	Electrical + mechanical	Heavy tow vehicle, level, hand tools

6.2 Installation Time Estimates

Solution	Setup Time	Equipment Connection	Commissioning	Total
WP-30	30 min	15 min	15 min	~1 hour
WP-60	45 min	15 min	15 min	~1.25 hours
WP-400T	1.5 hours	30 min	30 min	~2.5 hours
WP-900T	2 hours	45 min	30 min	~3.25 hours

6.3 Pre-Installation Checklist

- ☐ Site assessment completed (Section 4)
- ☐ Equipment manifest verified against delivery
- ☐ All components present and undamaged
- ☐ Battery state of charge verified (>40% recommended)
- ☐ Tools and PPE available
- ☐ Weather conditions acceptable (no rain, wind < 30 km/h)
- ☐ Communication with site supervisor confirmed

6.4 Solar Panel Orientation

For optimal energy harvest in Saudi Arabia:

- Panel orientation: SOUTH (0° azimuth)
- Tilt angle: 20-30° from horizontal (WP-60 adjustable 0-30°)
- Seasonal adjustment: 15° summer, 35° winter (if accessible)
- Avoid any shading, especially partial shading across cells

6.5 Trailer Setup Procedure (WP-400T, WP-900T)

- 1.** Position trailer on level ground with southern sky exposure
- 2.** Engage parking brake and chock wheels
- 3.** Deploy outriggers and level trailer (WP-400T)
- 4.** Disconnect from tow vehicle
- 5.** Deploy and orient solar panels
- 6.** Raise mast to desired height
- 7.** Mount equipment to mast/camera box
- 8.** Connect equipment cables
- 9.** Verify system operation

7. Solution Selection Decision Tree

Follow this flowchart to select the appropriate Power Solution:

[IMAGE: Decision tree flowchart: Start → Calculate Load → Select by load range → Verify site access → Confirm selection]

7.1 Decision Steps

1. Calculate TOTAL SYSTEM LOAD (not device power alone)
2. Check load against solution capacity table (Section 1.1)
3. Verify site access for selected solution
4. Confirm environmental compatibility
5. If multiple solutions qualify, prefer smaller for cost savings
6. If borderline, select larger solution for margin

7.2 Quick Decision Matrix

If your load is...	And site has...	Select...
≤8W (Gateway only)	Existing pole/wall	WP-30
9-15W (single camera)	Existing structure	WP-60
16-100W (PTZ or multi-camera)	Vehicle access + level ground	WP-400T
101-350W (surveillance system)	Heavy vehicle access + space	WP-900T
>350W	Any	Multiple units or custom solution

8. Common Deployment Scenarios

8.1 Scenario A: Remote Gateway

Situation: WakeCap Gateway for worker tracking at isolated site

- Equipment: WakeCap Gateway (~8W system)
- Solution: WP-30
- Mounting: Existing pole or temporary pole
- Installation: 1 technician, ~1 hour

8.2 Scenario B: Single Bullet Camera

Situation: Security camera monitoring equipment yard

- Equipment: Bullet camera + PoE + router (~35W system)
- Solution: WP-60 (15W continuous allows 24h backup at 35W)
- Alternative: WP-400T for full 48h backup
- Mounting: Bracket on structure or dedicated pole
- Installation: 1 technician, ~1.25 hours

8.3 Scenario C: Single PTZ Camera

Situation: PTZ surveillance camera at remote checkpoint

- Equipment: PTZ camera Type 1 (~70W system)
- Solution: WP-400T (100W capacity with margin)
- Mounting: Trailer mast (6.5m height)
- Installation: 2 technicians, ~2.5 hours

8.4 Scenario D: Multi-Camera Surveillance

Situation: Construction site perimeter monitoring

- Equipment: 2× PTZ cameras + 1× bullet + router (~200W system)
- Solution: WP-900T (350W capacity)
- Mounting: Trailer mast (9m height)
- Installation: 2-3 technicians, ~3.25 hours
- Note: Consider communication equipment power if cellular backhaul needed

8.5 Scenario E: High-Power PTZ System

Situation: High-end PTZ with heating element in harsh environment

- Equipment: PTZ Type 2 (~90W) + backup camera (~35W) = 125W system

- Solution: WP-900T
- Note: WP-400T insufficient for 48h backup at 125W

HYBRID DEPLOYMENTS

For some sites, deploying multiple smaller units may be more practical than one large unit. Example: Two WP-60 units for separate cameras vs. one WP-400T requires no trailer access but needs two mounting structures.

END OF DOCUMENT

WakeCap Power Solutions - Operations Selection Guide v1.0
For technical support: [TBD - Operations Support Contact]